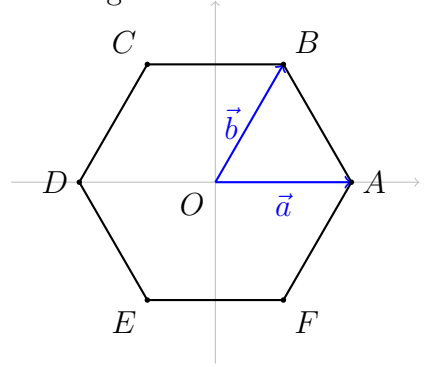


G11 PHYSICS

Assignment 01 Solutions

Problem 1. In the figure, A, B, C, D, E and F are vertices of a regular hexagon centered at the origin O . If $\vec{a} = \vec{OA}$ and $\vec{b} = \vec{OB}$, express each of the following in terms of $\vec{a}$ and $\vec{b}$:

- | | |
|----------------|--------------------------------------|
| (a) $\vec{AB}$ | (b) $\vec{BC}$ |
| (c) $\vec{AD}$ | (d) $\vec{CF}$ |
| (e) $\vec{AC}$ | (f) $\vec{BC} + \vec{DE} + \vec{FA}$ |



Solution: In a regular hexagon, triangles such as $\triangle OAB$ are equilateral, and opposite vertices lie on the same line through the center.

- (a) $\vec{OA} + \vec{AB} + \vec{BO} = \vec{0}$
 $\Rightarrow \vec{AB} = -\vec{OA} - \vec{BO} = -\vec{a} - (-\vec{b})$
 $= \vec{b} - \vec{a}$
- (b) $\vec{BC} = \vec{AO} = -\vec{a}$
- (c) $\vec{AD} = \vec{AO} + \vec{OD}$
 $= -\vec{a} + (-\vec{a}) = -2\vec{a}$
- (d) $\vec{OF} + \vec{FA} + \vec{AO} = \vec{0}$
 $\vec{OF} = -\vec{FA} - \vec{AO} = -\vec{b} - (-\vec{a}) = \vec{a} - \vec{b}$
 Then, $\vec{CF} = \vec{CO} + \vec{OF} = 2\vec{OF}$
 $= 2(\vec{a} - \vec{b})$
- (e) $\vec{AC} + \vec{CB} + \vec{BA} = \vec{0}$
 $\vec{AC} = -\vec{CB} - \vec{BA} = -\vec{CB} + \vec{AB} \quad (\because -\vec{BA} = \vec{AB})$
 $= -\vec{a} + (\vec{b} - \vec{a})$
 $= \vec{b} - 2\vec{a}$
- (f) $\vec{BC} + \vec{DE} + \vec{FA} = \vec{BC} + \vec{CO} + \vec{OB} = \vec{0}$

Problem 2. You are given three vectors:

$$\vec{a} = \hat{x} + 3\hat{y} - \hat{z}, \quad \vec{b} = 3\hat{x} - \hat{y} + \hat{z}, \quad \vec{c} = \hat{x} + 3\hat{y}$$

Find:

- (a) $\vec{a} - \vec{b}$ (c) $\vec{a} \cdot \hat{x}$ (e) $(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$
 (b) a_x (d) $\vec{a} \cdot \vec{b}$

Solution:

$$(a) \quad \vec{a} - \vec{b} = (1 - 3)\hat{x} + (3 - (-1))\hat{y} + (-1 - 1)\hat{z} = -2\hat{x} + 4\hat{y} - 2\hat{z}$$

$$(b) \quad a_x = 1$$

$$(c) \quad \vec{a} \cdot \hat{x} = (\hat{x} + 3\hat{y} - \hat{z}) \cdot \hat{x} = \hat{x} \cdot \hat{x} + 3\hat{y} \cdot \hat{x} - \hat{z} \cdot \hat{x} = 1$$

$$(d) \quad \vec{a} \cdot \vec{b} = (1)(3) + (3)(-1) + (-1)(1) = 3 - 3 - 1 = -1$$

$$\begin{aligned} (e) \quad (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} &= [(1)(1) + (3)(3) + (-1)(0)](3\hat{x} - \hat{y} + \hat{z}) - \\ &\quad [(1)(3) + (3)(-1) + (-1)(1)](\hat{x} + 3\hat{y}) \\ &= 10(3\hat{x} - \hat{y} + \hat{z}) + (\hat{x} + 3\hat{y}) \\ &= 31\hat{x} - 7\hat{y} + 10\hat{z} \end{aligned}$$

Problem 3. Two vectors having equal magnitude A make an angle θ with each other. Find the magnitude of the resultant and its direction relative to the first vector.

Solution: Suppose the two vectors be $\vec{A}$ and $\vec{B}$ such that

$$|\vec{A}| = |\vec{B}| = A,$$

and the angle between them is θ .

Let the resultant vector be

$$\vec{R} = \vec{A} + \vec{B}.$$

To find the magnitude of the resultant, consider

$$|\vec{R}|^2 = \vec{R} \cdot \vec{R}.$$

Substituting $\vec{R} = \vec{A} + \vec{B}$,

$$\begin{aligned} |\vec{R}|^2 &= (\vec{A} + \vec{B}) \cdot (\vec{A} + \vec{B}) = \vec{A} \cdot \vec{A} + 2\vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{B} \\ &= A^2 + A^2 + 2A^2 \cos \theta. \quad (\because \vec{A} \cdot \vec{A} = A^2, \vec{B} \cdot \vec{B} = A^2, \vec{A} \cdot \vec{B} = A^2 \cos \theta) \end{aligned}$$

Thus

$$R = \sqrt{2A^2(1 + \cos \theta)} = \sqrt{4A^2 \cos^2 \left(\frac{\theta}{2} \right)} \quad (\because 1 + \cos \theta = 2 \cos^2(\theta/2))$$

Hence

$$R = 2A \cos \left(\frac{\theta}{2} \right).$$

The resultant vector $\vec{R}$ will make an angle α with the first vector $\vec{A}$, where

$$\begin{aligned} \alpha &= \tan^{-1} \left(\frac{B \sin \theta}{A + B \cos \theta} \right) \\ &= \tan^{-1} \left(\frac{\sin \theta}{1 + \cos \theta} \right) \quad (\because A = B) \\ &= \tan^{-1} \left(\frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}} \right) \quad (\because \sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}, \text{ using the half-angle formula}) \\ &= \tan^{-1} \left(\tan \frac{\theta}{2} \right) \\ &= \frac{\theta}{2} \end{aligned}$$

Problem 4. Two vectors of equal magnitude 2 units have an angle 60° between them. Find the magnitude of:

- (a) The sum of the vectors.
- (b) The difference of the vectors.

Solution: Suppose the two vectors be $\vec{A}$ and $\vec{B}$ such that

$$|\vec{A}| = |\vec{B}| = 2,$$

and the angle between them is 60° .

(a) Let the resultant of the two vectors be

$$\vec{R} = \vec{A} + \vec{B}.$$

Then,

$$\begin{aligned} |\vec{R}|^2 &= \vec{R} \cdot \vec{R} \\ &= (\vec{A} + \vec{B}) \cdot (\vec{A} + \vec{B}) \quad (\because \vec{R} = \vec{A} + \vec{B}) \\ &= \vec{A} \cdot \vec{A} + 2\vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{B} \\ &= 4 + 2(2) + 4 \quad (\because \vec{A} \cdot \vec{A} = 4, \vec{B} \cdot \vec{B} = 4, \vec{A} \cdot \vec{B} = 2 \times 2 \cos 60^\circ = 2) \\ &= 12. \end{aligned}$$

Hence,

$$|\vec{R}| = \sqrt{12} = 2\sqrt{3} \text{ units}$$

(b) Let

$$\vec{D} = \vec{A} - \vec{B}.$$

Then

$$\begin{aligned} |\vec{D}|^2 &= (\vec{A} - \vec{B}) \cdot (\vec{A} - \vec{B}) \\ &= \vec{A} \cdot \vec{A} - 2\vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{B} \\ &= 4 - 4 + 4 \\ &= 4. \end{aligned}$$

Hence,

$$|\vec{D}| = 2 \text{ units.}$$

Problem 5. A force of 10 N acts on a particle along a direction making an angle 25° with the vertical. Find the component of the force in both the vertical and horizontal directions.

Solution:

Vertical component: $F_y = 10 \cos 25^\circ \approx 9.06 \text{ N}$

Horizontal component: $F_x = 10 \sin 25^\circ \approx 4.23 \text{ N}$

Problem 6. Find the separation vector $\vec{\Delta}$ from the source point $(3, 6, 9)$ to the field point $(0, 6, 8)$. Determine its magnitude (Δ) and construct the unit vector $\hat{\Delta}$.

Solution:

The position vectors of the source and field points are

$$\vec{r} = \langle 3, 6, 9 \rangle, \quad \vec{r}' = \langle 0, 6, 8 \rangle.$$

(Note: Remember $\vec{r} = (3 - 0)\hat{x} + (6 - 0)\hat{y} + (9 - 0)\hat{z} = 3\hat{x} + 6\hat{y} + 9\hat{z}$ (often represented as simply $\vec{r} = \langle 3, 6, 9 \rangle$ and same thing for $\vec{r}'$)

The separation vector from the source to the field point is

$$\begin{aligned} \vec{\Delta} &= \vec{r}' - \vec{r} \\ &= (0 - 3)\hat{x} + (6 - 6)\hat{y} + (8 - 9)\hat{z} \\ &= -3\hat{x} - \hat{z}. \end{aligned}$$

Magnitude

$$\begin{aligned} \Delta &= |\vec{\Delta}| \\ &= \sqrt{(-3)^2 + 0^2 + (-1)^2} \\ &= \sqrt{10}. \end{aligned}$$

Unit vector

$$\begin{aligned} \hat{\Delta} &= \frac{\vec{\Delta}}{|\vec{\Delta}|} \\ &= -\frac{1}{\sqrt{10}}(3\hat{x} + \hat{z}). \end{aligned}$$

Problem 7. A pilot flying at an air speed of 50 m s^{-1} wishes to travel due north. He knows that there is a 10 m s^{-1} wind blowing from west to east.

- (a) In what direction should he head his plane?
- (b) What will be the duration T of his flight if his destination is 160 km away?

Solution

Given: wind speed = 10 m s^{-1} blowing from west to east, plane air speed = 50 m s^{-1} , and distance = 160 km = 160,000 m.

(a) Since the pilot wishes to travel due north, the east–west components must cancel. Hence,

$$\sin \theta = \frac{\text{wind speed}}{\text{air speed}} = \frac{10}{50} = 0.2$$

Thus,

$$\theta = \sin^{-1}(0.2) \approx 11.5^\circ.$$

Therefore, the pilot must head 11.5° west of north.

(b) The northward component of the air speed is

$$\text{speed} = 50 \cos(11.5^\circ) \approx 49 \text{ m s}^{-1}.$$

The flight time T to reach the destination is

$$T = \frac{\text{distance}}{\text{speed}} \approx \frac{160000}{49}$$

$$\approx 3265.31 \text{ s}$$

$$T \approx 54.4 \text{ minutes}$$